Candidal and Dermatophyte Nail Involvement

Candidal onychomycosis is characterized by invasion of pseudohyphae throughout the entire nail plate, adjacent cuticle, granular layer, and stratum spinosum of the nail bed, as well as the hyponychial stratum corneum. In contrast, dermatophyte infections show a predilection for the ventral nail and nail bed stratum corneum. Histopathologically, the epidermis may exhibit spongiosis and focal parakeratosis, with minimal dermal inflammatory response. In white superficial onychomycosis (WSO), organisms are confined to the dorsal nail surface and display distinctive "perforating organs" and "eroding fronds."

Treatment of Dermatophytes

Tinea Capitis/Favus

Multiple systemic and topical antifungal agents are effective against dermatophytes. Infections involving hair-bearing skin require oral therapy due to fungal invasion of the hair follicle.

Griseofulvin

Griseofulvin remains the only oral antifungal approved by the U.S. Food and Drug Administration for tinea capitis. The traditional pediatric dose was 10–20 mg/kg/day for 6–8 weeks, administered with a fatty meal to enhance absorption. However, due to reported high failure rates, current expert recommendations favor higher doses: 20–25 mg/kg/day of the microsize formulation or 15 mg/kg/day of the ultramicrosize formulation, given for 8 weeks—though this is not based on controlled clinical trials.

Disadvantages include poor compliance due to treatment duration and cost, bitter taste in liquid form, photosensitivity, headache, and gastrointestinal side effects. Griseofulvin is a potent inducer of cytochrome P_{450} enzymes. Routine laboratory monitoring is not required for standard treatment courses.

Fluconazole

Available in both tablet and palatable liquid forms, fluconazole at 6 mg/kg/day for 20 days achieves an 89% cure rate in *Trichophyton tonsurans* tinea capitis. Alternative regimens include weekly dosing of 8 mg/kg for 8–16 weeks or 6 mg/kg daily for 2 weeks, followed by a single week of retreatment 4 weeks later if clinically indicated.

Fluconazole absorption is unaffected by food, and gastrointestinal side effects are uncommon. Hepatotoxicity has been reported but is less frequent than with ketoconazole.

Itraconazole

At doses of 3–5 mg/kg/day, itraconazole effectively clears tinea capitis caused by *Microsporum* or *Trichophyton* species within 4–6 weeks. Pulse therapy (5 mg/kg/day for 1 week per month, repeated for 1–3 cycles) is also effective.

Adverse effects include gastrointestinal upset, diarrhea (particularly with the liquid formulation), and peripheral edema—especially when co-administered with calcium channel blockers. Hepatotoxicity is rare compared to ketoconazole, though cases of hepatic failure have been reported. Rare associations with congestive heart failure exist.

Terbinafine

Terbinafine at 3–6 mg/kg/day cures *Trichophyton* tinea capitis in 2–4 weeks, but 4–8 weeks are needed for *Microsporum* infections. A randomized trial comparing 4 weeks of terbinafine to 8 weeks of griseofulvin showed no overall significant difference in efficacy. However, subgroup analysis revealed superior response to terbinafine in *Trichophyton* infections, whereas *M. audouinii* infections responded better to griseofulvin.